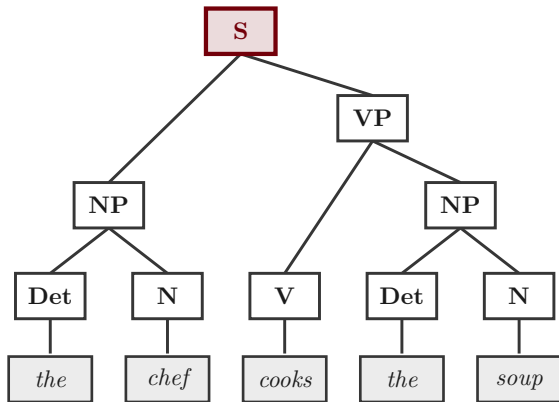


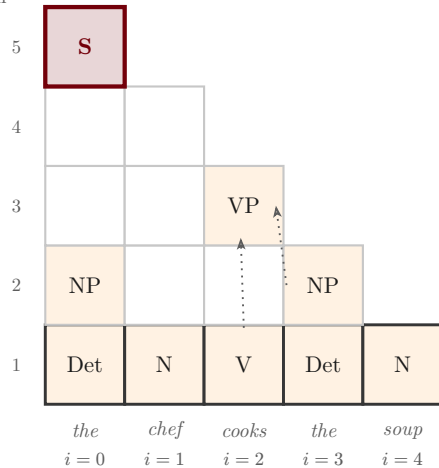
Constituency parse tree



phrase-structure derivation under a CFG

CKY span chart

span len $cell(i, j) = \text{non-terminals deriving } w_i \dots w_j$



$VP \rightarrow V \ NP$ fills [cooks] $\times$ [the soup]

top-right = S $\rightarrow$ sentence is in the language